

BARLAD, Romania

WMO: 151970

Lat: 46.2331N

Lon: 27.6444E

Elev: 173

StdP: 99.26

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-11.1	-16.2	0.9	-11.7	-13.6	1.2	-9.5	12.3	-2.9	9.9	-6.1	3.0	0	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.3	21.1	30.9	20.5	29.6	20.2	22.9	29.7	22.0	28.1	21.1	27.3	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.7	25.6	19.9	14.9	24.6	19.1	14.2	23.6	68.3	29.7	65.0	28.3	62.0	27.2	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.3	DB	-16.4	34.5	3.0	1.8	-18.6	35.8	-20.3	36.8	-22.0	37.8	-24.2	39.1	(4)
(5)			WB	-16.9	24.3	3.3	1.0	-19.3	25.0	-21.2	25.5	-23.1	26.1	-25.5	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.2	1.3	5.1	11.2	16.6	20.9	22.1	22.0	17.3	10.9	4.8	-0.9	(6)	
	DBStd	9.61	4.36	5.16	4.76	4.26	3.51	2.79	2.59	3.21	4.22	4.22	4.92	5.03	(7)	
	HDD10.0	1373	378	246	161	37	3	0	0	0	1	40	168	339	(8)	
	HDD18.3	3141	637	478	411	216	75	9	3	7	69	233	406	597	(9)	
	CDD10.0	1662	0	1	8	73	206	327	376	371	220	67	12	0	(10)	
	CDD18.3	388	0	0	0	1	20	86	121	120	38	2	0	0	(11)	
	CDH23.3	3649	0	0	1	18	225	751	1099	1182	361	12	0	0	(12)	
	CDH26.7	1206	0	0	0	2	42	216	381	465	100	1	0	0	(13)	
(14) Wind	WSAvg	2.9	3.1	3.3	3.8	3.4	2.9	2.7	2.6	2.5	2.7	2.7	2.9	3.0	(14)	
(15) Precipitation	PrecAvg	511	22	21	28	45	58	71	54	50	43	50	32	33	(15)	
	PrecMax	693	45	68	59	86	158	161	151	157	123	197	104	109	(16)	
	PrecMin	312	1	0	3	12	4	18	8	13	4	3	1	3	(17)	
	PrecStd	118	14	16	16	21	35	39	33	34	29	40	24	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	16.2	20.7	25.4	30.0	32.4	33.5	34.5	31.2	25.0	19.3	12.0	(19)	
		MCWB	6.1	11.0	12.0	16.5	19.4	22.4	22.0	22.3	18.8	17.2	13.6	9.3	(20)	
	2%	DB	6.3	12.6	16.7	23.0	27.6	30.5	31.9	32.8	29.5	22.0	16.2	9.5	(21)	
		MCWB	4.9	8.8	10.1	14.6	18.3	21.2	21.7	20.7	18.8	15.3	12.2	7.8	(22)	
	5%	DB	4.7	9.9	14.4	20.7	25.9	29.2	30.5	31.1	27.5	19.8	13.5	7.4	(23)	
		MCWB	3.7	7.1	9.0	13.6	17.9	20.7	21.0	20.0	17.6	14.3	10.7	5.9	(24)	
	10%	DB	3.2	7.8	12.2	18.7	23.9	27.7	29.0	29.4	25.2	17.5	11.6	5.0	(25)	
		MCWB	2.4	5.6	7.7	12.6	16.7	20.1	20.3	19.6	17.5	13.1	9.4	3.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	11.8	12.5	17.0	20.9	23.8	23.8	24.1	21.1	18.4	14.4	10.5	(27)	
		MCDB	8.3	16.2	19.1	24.4	28.6	30.1	30.5	32.5	27.6	22.5	17.7	11.2	(28)	
	2%	WB	5.2	9.1	10.6	15.4	19.6	22.6	22.7	22.5	20.2	16.2	12.7	8.0	(29)	
		MCDB	6.0	11.8	15.4	21.9	26.3	28.7	29.4	29.6	25.8	20.9	15.7	9.6	(30)	
	5%	WB	4.0	7.4	9.4	14.2	18.5	21.7	21.9	21.4	19.2	15.0	11.4	5.9	(31)	
		MCDB	4.8	9.4	13.2	19.5	24.4	27.3	28.3	28.0	24.1	18.9	13.6	7.4	(32)	
	10%	WB	2.6	6.0	8.2	13.1	17.3	20.8	21.0	20.5	18.2	13.7	9.7	3.9	(33)	
		MCDB	3.4	7.4	11.3	17.9	22.4	26.4	27.1	27.0	22.9	16.6	11.4	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.8	8.5	10.0	10.9	11.0	11.1	11.7	10.8	8.8	6.1	5.3	(35)	
		MCDBR	6.9	10.6	12.6	13.9	14.0	13.1	13.6	14.4	14.1	12.3	9.7	7.4	(36)	
	5% WB	MCWBR	5.2	7.2	7.5	6.9	6.4	5.4	5.2	5.0	5.5	6.1	5.5	5.9	(37)	
		MCWBR	6.2	10.4	11.6	12.4	13.0	11.9	11.6	12.8	11.7	11.1	8.6	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.358	0.394	0.452	0.423	0.421	0.443	0.454	0.412	0.379	0.353	0.326	(40)	
		taud	2.284	2.238	2.198	2.060	2.223	2.264	2.210	2.169	2.249	2.321	2.337	2.312	(41)	
	Ebn at Noon		749	798	822	804	845	848	822	794	795	767	714	705	(42)	
		Edh at Noon	80	103	124	155	137	132	137	138	117	94	77	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.26	2.05	3.24	4.43	5.67	5.96	6.03	5.47	3.92	2.52	1.34	1.03	(44)	
	RadStd		0.17	0.26	0.32	0.41	0.48	0.50	0.47	0.39	0.39	0.28	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+0.63	N/A	N/A	+0.69	N/A	N/A	-137	-183	+71	+17

Nomenclature: See separate page